

CS 162 Homework Test Completeness Audit

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Scope

This report validates the homework documentation and the current testing state of this repository. It is based on the public CS 162 homework pages, the files in this repository, and the commands that could be attempted in the local Windows environment.

Official sources used:

- <<https://cs162.org/static/hw/hw-intro/>>
- <<https://cs162.org/static/hw/hw-list/>>
- <<https://cs162.org/static/hw/hw-shell/>>
- <<https://cs162.org/static/hw/hw-http/>>
- <<https://cs162.org/static/hw/hw-http-rs/>>
- <<https://cs162.org/static/hw/hw-memory/>>
- <<https://cs162.org/static/hw/hw-map-reduce/>>
- <<https://cs162.org/static/hw/hw-map-reduce-rs/>>

Environment validation

The local environment is useful for static review, but it is not sufficient for full CS 162 execution testing.

Attempted commands:

Check	Result	Impact	
make --version	Blocked: make is not installed	C homework builds and Pintos tests cannot be executed locally.	
cargo --version	Blocked: cargo is not installed	Rust homework tests cannot be executed locally.	
bash -lc "uname -a"	Blocked: WSL reports no installed Linux distribution	Linux shell scripts and Unix signal tests cannot be executed locally.	
rg -n "TODO	todo!" ...	Passed	Static implementation gaps were found and recorded below.
rg -n "[\p{Han}]" README.md hw-*/README.md	Passed earlier	Repository homework documentation is English-only.	
git diff --check	Passed earlier	Markdown files do not contain whitespace errors.	

Static evidence summary

The repository is not ready to be called fully tested. Static inspection found 53 TODO or todo! markers across homework code and documentation. The most important markers are implementation placeholders, not just comments.

Observed implementation gap counts by homework:

```
| Homework | Static TODO markers | Interpretation |
| --- | ---: | --- |
| hw-intro | 0 | No obvious placeholder markers, but no automated tests were found. |
| hw-list | 13 | List and pthread word-count code still has visible TODO markers. |
| hw-shell | 0 | No TODO markers, but the test harness is only a smoke test. |
| hw-http | 15 | The C HTTP server still has several part-level TODO markers. |
| hw-http-rs | 6 | The Rust HTTP server still has todo! placeholders in core modules. |
| hw-memory | 3 | The standalone allocator has unimplemented allocation functions. |
| hw-map-reduce | 6 | The C coordinator has visible TODO markers. |
| hw-map-reduce-rs | 10 | The Rust coordinator and ignored test still contain placeholders.
|
```

Existing test inventory

```
| Homework | Existing tests or harness | Completeness verdict |
| --- | --- | --- |
| hw-intro | No checked-in automated harness found | Not complete |
| hw-list | No checked-in automated harness found | Not complete |
| hw-shell | test.sh checks one pwd case and one /bin/lis case | Not complete |
| hw-http | No checked-in automated HTTP harness found | Not complete |
| hw-http-rs | 12 helper-level Rust tests in src/tests | Partial, not complete |
| hw-memory | mm_alloc/mm_test.c; 26 active Pintos memory tests in Make.tests | Partial;
Pintos side is strongest, standalone allocator is not complete |
| hw-map-reduce | No checked-in automated cluster harness found | Not complete |
| hw-map-reduce-rs | One ignored test_wc containing todo!("Pick an input file") | Not
complete |
```

Homework-by-homework validation

HW 0: Intro

Local execution status: blocked by missing make.

Documentation correctness check:

- The README points to the correct official HW 0 page and counting-words task.
- The listed files match the repository structure.
- The missing-test prompts cover word totals, frequency output, stdin, multiple files, invalid flags, limits, and map output shape.

Completeness verdict: not complete.

Required next tests:

- Add a words/tests/ fixture directory.
- Add golden-output tests for ./words -c and ./words -f.
- Add stdin and multi-file tests.
- Add shape checks for limits and map.

Minimum command set to run in Linux or WSL:

```
cd hw-intro
make
```

```
./limits
./map
cd words
make
./words -c tests/basic.txt
./words -f tests/basic.txt
```

HW 1: Lists

Local execution status: blocked by missing make.

Static evidence:

- word_count_l.c, word_count_p.c, and pwords.c still contain TODO markers.

Documentation correctness check:

- The README maps the official list and pwords tasks to list.c, word_count_l.c, word_count_p.c, and pwords.c.
- The recommended tests compare words, lwords, and pwords, which is the right local strategy for this homework.

Completeness verdict: not complete.

Required next tests:

- Add deterministic word-count fixtures.
- Compare lwords and pwords against the reference words binary.
- Add a repeated pwords race test.
- Add memory-check instructions with Valgrind or sanitizers.

Minimum command set to run in Linux or WSL:

```
cd hw-list
make
./words tests/fixtures/single.txt > /tmp/ref.out
./lwords tests/fixtures/single.txt > /tmp/lwords.out
./pwords tests/fixtures/single.txt > /tmp/pwords.out
diff -u /tmp/ref.out /tmp/lwords.out
diff -u /tmp/ref.out /tmp/pwords.out
```

HW 2: Shell

Local execution status: blocked by missing make and unavailable WSL Linux environment.

Static evidence:

- test.sh exists and is readable.
- It currently checks only pwd and /bin/ls.

Documentation correctness check:

- The README covers the official shell surfaces: directory commands,

foreground execution, redirection, pipes, Ctrl-D, Ctrl-C, and Ctrl-Z.

- The missing-test prompts include both deterministic command tests and environment-sensitive signal tests.

Completeness verdict: not complete.

Required next tests:

- Expand test.sh into a table-driven harness.
- Add cd, invalid command, redirection, pipe, and tokenizer tests.
- Add Linux-only signal tests for Ctrl-C and Ctrl-Z behavior.

Minimum command set to run in Linux or WSL:

```
cd hw-shell
make
bash test.sh
printf "cd /npwd\n" | ./shell
printf "echo hello | wc -w\n" | ./shell
```

HW 3: HTTP Server in C

Local execution status: blocked by missing make.

Static evidence:

- httpserver.c contains part-level TODO markers for file serving, directory serving, proxying, and concurrency paths.
- No checked-in HTTP integration test script was found.

Documentation correctness check:

- The README maps the official socket and HTTP tasks to httpserver.c, libhttp.c, wq.c, and the server variants.
- The missing-test prompts require real socket tests, not just unit tests.

Completeness verdict: not complete.

Required next tests:

- Add tests/http_smoke.sh.
- Run it against httpserver, forkserver, threadserver, and poolserver.
- Add status, header, body, directory, traversal, malformed request, proxy, and concurrency checks.

Minimum command set to run in Linux or WSL:

```
cd hw-http
make
./httpserver --files www --port 8000
curl -i http://127.0.0.1:8000/
curl -i http://127.0.0.1:8000/not-found
```

HW 3: HTTP Server in Rust

Local execution status: blocked by missing cargo.

Static evidence:

- src/http.rs, src/server.rs, and src/stats.rs contain todo! placeholders.
- src/tests contains 12 helper-level tests.
- No integration test starts the real server and sends TCP requests.

Documentation correctness check:

- The README correctly separates helper-level testing from socket-level integration testing.
- The missing-test prompts call for parser tests, real GET tests, directory tests, stats integration checks, and concurrent client coverage.

Completeness verdict: partial, not complete.

Required next tests:

- Keep current helper tests.
- Add tests/server_integration.rs.
- Add request parser coverage and real TCP client tests.
- Verify stats after real requests.

Minimum command set to run in Linux, WSL, or a Rust environment:

```
cd hw-http-rs
cargo test
cargo test --test server_integration
cargo run -- --files www --port 8000
```

HW 4: Memory

Local execution status: blocked by missing make and unavailable Pintos Linux environment.

Static evidence:

- mm_alloc/mm_alloc.c contains TODO markers for malloc, realloc, and free.
- pintos/src/tests/memory/Make.tests enables 26 active memory tests.
- Stack-growth tests are present but commented out.

Documentation correctness check:

- The README correctly distinguishes the standalone allocator from the Pintos memory suite.
- It avoids claiming active calloc coverage where the current active memory test list does not show explicit calloc tests.

Completeness verdict: partial, not complete.

Required next tests:

- Expand standalone allocator tests for alignment, reuse, realloc growth and

shrink, fragmentation, and stress.

- Run Pintos make check in the course environment.
- Decide whether stack-growth tests are required for the target assignment version before enabling them.

Minimum command set to run in Linux or the course VM:

```
cd hw-memory/mm_alloc
make
./mm_test
```

```
cd ../pintos/src/memory
make
make check
```

HW 5: MapReduce in C

Local execution status: blocked by missing make.

Static evidence:

- The C coordinator contains TODO markers.
- No checked-in integration harness was found.

Documentation correctness check:

- The README maps the official job-submission and cluster behavior to the RPC, coordinator, worker, client, and app directories.
- The missing-test prompts require a real cluster test with cleanup and golden output comparison.

Completeness verdict: not complete.

Required next tests:

- Add tests/run_cluster_test.sh.
- Start coordinator and workers automatically.
- Submit wc, grep, and vertex_degree jobs.
- Compare processed output against golden files.
- Add worker-failure retry coverage.

Minimum command set to run in Linux or WSL:

```
cd hw-map-reduce
make
./bin/mr-coordinator
./bin/mr-worker
./bin/mr-client submit -a wc -o /tmp/hw-map-reduce-out -w data/gutenberg/p.txt
./bin/mr-client process -a wc -o /tmp/hw-map-reduce-out
```

HW 5: MapReduce in Rust

Local execution status: blocked by missing cargo.

Static evidence:

- src/coordinator/mod.rs still contains job-submission TODOs.
- src/tests/mod.rs::test_wc is ignored and contains todo!("Pick an input file").

Documentation correctness check:

- The README covers worker registration, job submission, task assignment, completion, heartbeat, retry, and end-to-end output validation.
- The missing-test prompts ask for both state-machine unit tests and async integration tests.

Completeness verdict: not complete.

Required next tests:

- Finish test_wc with a real fixture and expected output.
- Add worker-registration, job-submission, task-assignment, completion, and heartbeat/retry tests.
- Add wc, grep, and vertex_degree integration tests with timeouts.

Minimum command set to run in Linux, WSL, or a Rust environment:

```
cd hw-map-reduce-rs
cargo test
cargo test test_wc -- --ignored
```

Overall verdict

The homework documentation is now stronger than a plain official assignment page because it is tied directly to this repository and its current testing state. However, the assignments themselves are not fully tested yet.

Readiness by homework:

Homework Documentation readiness Test completeness
--- --- ---
HW 0 Intro Good Not complete
HW 1 Lists Good Not complete
HW 2 Shell Good Not complete
HW 3 HTTP C Good Not complete
HW 3 HTTP Rust Good Partial, not complete
HW 4 Memory Good Partial, not complete
HW 5 MapReduce C Good Not complete
HW 5 MapReduce Rust Good Not complete

Definition of done for future work

Each homework should be considered well tested only after all of the following are true:

1. The homework builds in the expected Linux, WSL, or course VM environment.
2. The documented smoke commands pass.

3. The repository contains deterministic automated tests for the main required behavior.
4. Edge cases and failure modes listed in each README are covered.
5. Long-running or environment-sensitive tests have timeouts.
6. The test commands are documented and reproducible from a fresh checkout.
7. Any skipped or ignored test is explained in the README or audit report.